

Education

Tsinghua University

Beijing, China

Ph.D., Software Engineering

Aug 2020 - Dec 2025

Thesis: Research on Backscatter Communication Technology for Large-Scale Battery-Free Sensing Network

Advisor: Prof. [Yuan He](#)

With honor: *Talents Scholarships* in Tsinghua University (*Top 15%*)

Tsinghua University

Beijing, China

B.E., Software Engineering

Aug 2016 - Jul 2020

With honor: *Excellent Graduate* in Tsinghua University and *Outstanding Graduate* in School of Software.

Research Interest

Wireless Networking: Design and implementation of energy-efficient wireless communication systems and networking architectures for scalable Internet-of-Things applications, including low-power wireless designs and cross-technology communication for smart homes, industrial IoT, and human-centered environments.

Ubiquitous Sensing: Development of low-power and battery-free sensing platforms for continuous monitoring in everyday and industrial environments, with an emphasis on scalable, sustainable, and application-oriented IoT solutions.

Honors and Awards

Best Paper Nominee Award (<i>2 out of 233 submissions</i>), ACM MobiSys	Jun 2025
Talents Scholarship (<i>Top 15%</i>), Tsinghua University	2021 - 2024
Excellent University Graduate (<i>Top 10%</i>), Tsinghua University	Jun 2020
Outstanding Graduate (<i>Top 10%</i>), School of Software	Jun 2020
National Encouragement Scholarship (<i>Top 5%</i>), Tsinghua University	2018, 2019

Publications

[1] Analog Backscatter for Commodity WiFi with Payload Transparency

Xin Na, Yuan He, Xiuzhen Guo, Jia Zhang, Yang Zou, Zihao Yu, Yunhao Liu.

IEEE/ACM Transactions on Networking (ToN), 2025. [[PDF](#)]

[2] QuinID: Enabling FDMA-Based Fully Parallel RFID with Frequency-Selective Antenna

Xin Na, Jia Zhang, Jiacheng Zhang, Xiuzhen Guo, Yang Zou, Meng Jin, Yimiao Sun, Yunhao Liu, Yuan He.

ACM International Conference on Mobile Computing and Networking (MobiCom), 2025. [[PDF](#)] [[Code](#)] [[Slides](#)]

[3] Leggiero: Analog WiFi Backscatter with Payload Transparency

Xin Na, Xiuzhen Guo, Zihao Yu, Jia Zhang, Yuan He, Yunhao Liu.

ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2023. [[PDF](#)] [[Code](#)] [[Slides](#)]

[4] Wi-attack: Cross-technology Impersonation Attack against iBeacon Services

Xin Na, Xiuzhen Guo, Yuan He, Rui Xi.

IEEE International Conference on Sensing, Communication, and Networking (SECON), 2021. [[PDF](#)] [[Code](#)] [[Slides](#)]

[5] Satori: In-band Analog Backscatter for Audio Transmission

Yang Zou, **Xin Na**, Yimiao Sun, Yande Chen, Yuan He.

ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025. [[PDF](#)]

Best Paper Nominee, 2 out of 233.

[6] SmarTiSCH: An Interference-Aware Engine for IEEE 802.15.4e-based Networks

Zihao Yu, **Xin Na**, Carlo Alberto Boano, Yuan He*, Xiuzhen Guo, Pengyu Li, Meng Jin.

ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN), 2022. [[PDF](#)]

- [7] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division
Yang Zou, **Xin Na**, Xiuzhen Guo, Yimiao Sun, Yuan He.
IEEE International Conference on Computer Communications (INFOCOM), 2024. [PDF]
- [8] mmHawkeye: Passive UAV Detection with a COTS mmWave Radar
Jia Zhang, **Xin Na**, Rui Xi, Yimiao Sun, Yuan He.
IEEE International Conference on Sensing, Communication, and Networking (SECON), 2023. [PDF]
- [9] Trident: Interference Avoidance in Multi-reader Backscatter Network via Frequency-space Division
Yang Zou, **Xin Na**, Yimiao Sun, Yuan He.
IEEE/ACM Transactions on Networking (ToN), 2024. [PDF]
- [10] BIFROST: Reinventing WiFi Signals Based on Dispersion Effect for Accurate Indoor Localization
Yimiao Sun, Yuan He, Jiacheng Zhang, **Xin Na**, Yande Chen, Weiguo Wang, Xiuzhen Guo.
ACM Conference on Embedded Networked Sensor Systems (SenSys), 2023. [PDF]
- [11] Detection and Identification of Non-cooperative UAV Using a COTS mmWave Radar
Yuan He, Jia Zhang, Rui Xi, **Xin Na**, Yimiao Sun, Beibei Li.
ACM Transactions on Sensor Networks (TOSN), 2024.[PDF]
- [12] RFinder: Pinpoint the Invisible RFID Tags in the Prefabricated Buildings
Jiacheng Zhang, Meng Jin, Yimiao Sun, Weiguo Wang, Jia Zhang, **Xin Na**, Xiuzhen Guo, Yuan He.
International Conference on Embedded Wireless Systems and Networks (EWSN), 2024.[PDF]
- [13] Taming the Errors in Cross-Technology Communication: A Probabilistic Approach
Xiuzhen Guo, Yuan He, Jia Zhang, Haotian Jiang, Zihao Yu, **Xin Na**.
ACM Transactions on Sensor Networks (TOSN), 2021.[PDF]
- [14] Real-Time Continuous Activity Recognition With a Commercial mmWave Radar
Yunhao Liu, Jia Zhang, Yande Chen, Weiguo Wang, Songzhou Yang, **Xin Na**, Yimiao Sun, Yuan He.
IEEE Transactions on Mobile Computing (TMC), 2024.[PDF]
- [15] Cross-Technology Communication for the Internet of Things: A Survey
Yuan He, Xiuzhen Guo, Xiaolong Zheng, Zihao Yu, Jia Zhang, Haotian Jiang, **Xin Na**, Jiacheng Zhang.
ACM Computing Surveys (CSUR), 2022.[PDF]
- [16] A Survey of mmWave-Based Human Sensing: Technology, Platforms and Applications
Jia Zhang, Rui Xi, Yuan He, Yimiao Sun, Xiuzhen Guo, Weiguo Wang, **Xin Na**, Yunhao Liu, Zhenguo Shi, Tao Gu.
IEEE Communications Surveys & Tutorials (COMST), 2023.[PDF]

Tools and Open-Source Researches

Beyond research, I also dedicate myself to building and sharing tools, hardware, and software to advance the wireless networking community.

[QuinID](#), hardware and software for an FDMA-based fully parallel RFID system.

- The UHF RFID band is divided into five sub-bands, resulting in a fivefold improvement in reading rate.
- An additional tool, [UHF-Gen2-RFID-Reader](#), is provided—a high-performance SDR-based reader for the broad RFID research community, achieving up to 1000 reads per second.

[Leggiero](#), the entire hardware of an analog WiFi backscatter.

- The tag converts the analog sensor voltage to the RF signal phase and embeds it in the CSI.
- An updated version, [Leggiero+](#), is also provided with a few enhancements made to the circuit design.

[Wi-attack](#), the software of a cross-technology impersonation attack method against iBeacon.

- Uses WiFi devices to attack iBeacon services by impersonating an iBeacon base station.

Professional Activities

IEEE Journal of Radio Frequency Identification, Reviewer	2026 - Now
ACM Transactions on Internet of Things, Distinguished Reviewer	2025 - Now
IEEE Transactions on Mobile Computing, Reviewer	2024 - Now
Contributed to paper reviews for INFOCOM, IPSN, SenSys, SECON	2021 - Now
Teaching Assistant, Network Measurement and Analysis, Instructor: Prof. Yuan He	2021 - 2024